Properties of powers with a natural exponent

Five laws — and every one of them is just counting the factors.

LESSON 63–64 | 2 × 40 MIN | MATEMATIKA 7, §28 | S8 6.2

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

Multiplying and dividing	15 min
Powers of powers	25 min
Products and quotients	25 min
Putting them together	20 min
Homework	5 min

LEARNING OBJECTIVES

- State and prove the laws of indices for natural exponents.
- Multiply and divide powers with the same base.
- Raise a power, a product and a quotient to a power.
- Simplify an expression containing several powers.

TEXTBOOK

National	pp. 171–177
Cambridge	Stage 8, pp. 60–65
Workbook	Exercise 6.2

01 Explanation

Multiplying and dividing

$$a^m \cdot a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \ (m > n)$$

Why. $a^3 \cdot a^2 = (aaa)(aa) = a^5$: the exponents count the factors, and multiplying puts the two lists end to end.

EXPRESSION	WORKING	ANSWER
$2^3 \cdot 2^4$	2^{3+4}	$2^7 = 128$
$x^5 \cdot x$	x^{5+1}	x^6
$3^7 \div 3^4$	3^{7-4}	$3^3 = 27$
$a^6 \div a^6$	a^0	1

THE BASES MUST BE THE SAME

$2^3 \cdot 3^2$ cannot be combined — the laws say nothing about it. Only $2^3 \cdot 2^2$ becomes 2^5.

Powers of powers

$$(a^m)^n = a^{mn}$$

Why. $(a^3)^2 = a^3 \cdot a^3 = a^6$: n copies of m factors is mn factors.

EXPRESSION	ANSWER
$(2^3)^2$	$2^6 = 64$
$(x^4)^3$	x^{12}
$(a^2)^5$	a^{10}

$(A^3)^2$ AND $A^3 \cdot A^2$ ARE DIFFERENT

The first is a^6, the second a^5. Multiply the exponents for a power of a power; add them for a product.

Products and quotients

$$(ab)^n = a^n b^n \quad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

EXPRESSION	ANSWER
$(2x)^3$	$8x^3$
$(-3a)^2$	$9a^2$
$\left(\frac{x}{2}\right)^4$	$\frac{x^4}{16}$
$(2 \times 5)^3$	1000

THERE IS NO SUCH LAW FOR A SUM

$(a + b)^2 \neq a^2 + b^2$. Try $a = b = 1$: $4 \neq 2$. The correct expansion is the subject of Quarter III.

Putting them together

EXPRESSION	WORKING	ANSWER
$(2x^2)^3$	$2^3(x^2)^3$	$8x^6$
$x^5 \cdot x^3 \div x^6$	x^{8-6}	x^2
$(a^3b)^2 \cdot a$	$a^6b^2 \cdot a$	a^7b^2
$\frac{(3a^2)^3}{9a^4}$	$\frac{27a^6}{9a^4}$	$3a^2$

DEAL WITH EACH BASE SEPARATELY

Collect the numbers, then the as, then the bs. Trying to do all three at once is where the errors come from.

02 Worked examples

EXAMPLE 1 **Simplify $2^3 \cdot 2^4$, $3^7 \div 3^4$ and $(2^3)^2$.**

1. Same base, multiply: add the exponents.
2. $2^7 = 128$
3. Same base, divide: subtract.

   $3^3 = 27$.
4. Power of a power: multiply. $2^6 = 64$.

ANSWER

$128, 27, 64$

EXAMPLE 2 **Simplify $(2x^2)^3$.**

1. Raise each factor: $2^3 \cdot (x^2)^3$.
2. $2^3 = 8$
3. $(x^2)^3 = x^6$

   Multiply the exponents.
4. $= 8x^6$

ANSWER

$8x^6$

EXAMPLE 3 **Simplify $\frac{(3a^2)^3}{9a^4}$.**

1. Numerator: $27a^6$.
2. Numbers: $\frac{27}{9} = 3$.
3. Letters: $a^6 \div a^4 = a^2$.
4. $= 3a^2$

ANSWER

$3a^2$

03 Interactive model

Write a³ · a² out in full as aaaaa before stating the law; the class sees that the rule is only counting.

Combining powers

$$\sqrt[3]{64} = 64^{1/3} = \mathbf{4}$$

base a **64** | index n **3** | power m **1** | check: value^3 **64** | a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Law of indices	Daraja qonuni	Свойство степени
Same base	Bir xil asos	Одинаковое основание
To multiply powers	Darajalarni ko‘paytirish	Умножение степеней
To divide powers	Darajalarni bo‘lish	Деление степеней
Power of a power	Daraja darajasi	Степень степени
Power of a product	Ko‘paytma darajasi	Степень произведения
Exponent zero	Nolinchi daraja	Нулевая степень
Simplify	Soddalashtirish	Упростить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$a^m \cdot a^n$ equals:

- a^{mn}
- a^{m+n}
- a^{m-n}
- $2a^m$

$(a^m)^n$ equals:

- a^{m+n}
- a^{mn}
- a^{m-n}
- a^m

$3^7 \div 3^4$ equals:

- 3^3
- 3^{11}
- 3^{28}
- 1

$(2x)^3$ equals:

- $2x^3$
- $6x^3$
- $8x^3$
- $8x$

$(a + b)^2$ equals:

- $a^2 + b^2$
- $a^2 + 2ab + b^2$
- $2a + 2b$
- ab

$a^6 \div a^6$ equals:

- 0
- 1
- a
- a^{12}

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $2^3 \cdot 2^4$

ANSWER 2^7

2. $x^5 \cdot x$

ANSWER x^6

3. $3^7 \div 3^4$

ANSWER 3^3

4. $(2^3)^2$

ANSWER 2^6

5. $(x^4)^3$

ANSWER x^{12}

6. $(2x)^3$

ANSWER $8x^3$

7. $a^6 \div a^6$

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. $(2x^2)^3$

ANSWER $8x^6$

2. $(-3a)^2$

ANSWER $9a^2$

3. $\left(\frac{x}{2}\right)^4$

ANSWER $\frac{x^4}{16}$

4. $x^5 \cdot x^3 \div x^6$

ANSWER x^2

5. $(a^3b)^2 \cdot a$

ANSWER a^7b^2

6. $\frac{(3a^2)^3}{9a^4}$

ANSWER $3a^2$

7. $2^3 \cdot 3^2$

ANSWER 72 — bases differ

Hard · stretch and reason

7 PROBLEMS

1. $\dfrac{(2a^2b)^3}{4a^3b}$

   ANSWER $2a^3b^2$

2. $(x^2y^3)^2 \cdot (xy)^3$

   ANSWER x^7y^9

3. $\dfrac{5^{n+1}}{5^n}$

   ANSWER 5

4. $(-2a^2)^3 \div (4a^3)$

   ANSWER $-2a^3$

5. Is $(a + b)^2 = a^2 + b^2$?

   ANSWER No — try $a = b = 1$

6. $2^{10} \div 2^7$

   ANSWER 8

7. $(3x)^2 \cdot (2x)^3$

   ANSWER $72x^5$

07 Homework

Homework — 5 tasks

Deal with the numbers first, then each letter separately.

1. Simplify $5^2 \cdot 5^3$, $7^6 \div 7^4$ and $(4^2)^3$.
2. Simplify $(3x^3)^2$ and $(-2a)^4$.
3. Simplify $y^7 \cdot y^2 \div y^5$.
4. Simplify $\dfrac{(2a^3)^4}{8a^5}$.

5. Show with numbers that $(a + b)^2 \neq a^2 + b^2$.